LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Miscellaneous loading cases, longitudinal cases, fuselage pressurization loads, flap and airbrake loads, spinning and power plants loads, fuel systems, freight loads, ground handling guided missile launcher cases.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Miscellaneous loading cases, longitudinal cases, fuselage pressurization loads, flap and airbrake loads, spinning and power plants loads, fuel systems, freight loads, ground handling guided missile launcher cases.
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of loading case diagrams from IB; Instructor supervision on identifying load paths.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary forces acting on an aircraft structure?
(b) Define the term 'loading case' in the context of aircraft design.
(c) Why is understanding fuselage pressurization loads critical for crew safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Longitudinal loading cases definition • Fuselage pressurization load mechanisms • Flap and airbrake load distribution • Spinning and power plant loads • Fuel system and freight load effects	L & Ds	PPT, IB	15 min
(b)	• Ground handling guided missile launcher cases • Comparison of miscellaneous loading scenarios • Impact of structural elements on load transfer • Analysis of load paths in different cases • Safety implications of excessive loading	L & Ds	PPT, PDF	20 min
(c)	• Summary of key loading case characteristics • Identification of critical structural elements • Review of fuel and freight load constraints • Reinforcement of pressurization load concepts • Preparation for next lesson on joining techniques	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how flap loads differ from airbrake loads in terms of structural stress.
(b) Describe the role of the fuselage in managing pressurization loads during flight.
(c) How do ground handling loads for guided missile launchers compare to standard freight loads?

12. Home Assignment and Follow Up	Prepare a brief note on the structural elements discussed in this lesson and how they relate to the classification of aircraft parts covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Classification of aircraft parts, types of aircraft fuselage structure, joining techniques, riveted joints, bonded joints, brazing.	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min